

PROJECT AND TEAM INFORMATION

Project Title

Personalized Film Recommendation System

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Online movie platforms contain a very large number of movies. A user may know that they enjoyed one movie but may not know which other movies are similar. Manually searching through thousands of titles is time-consuming. Therefore, there is a need for a system that analyzes movie information and automatically recommends relevant movies.

State of the Art / Current solution

- To collect and process movie metadata such as titles, genres, keywords, cast, director, and descriptions.
- To convert textual movie information into numerical representations using TF-IDF.
- To calculate similarity between movies using Cosine Similarity.
- To recommend the top similar movies for a selected movie.
- To provide an easy-to-use interface for demonstrating the recommendation system.

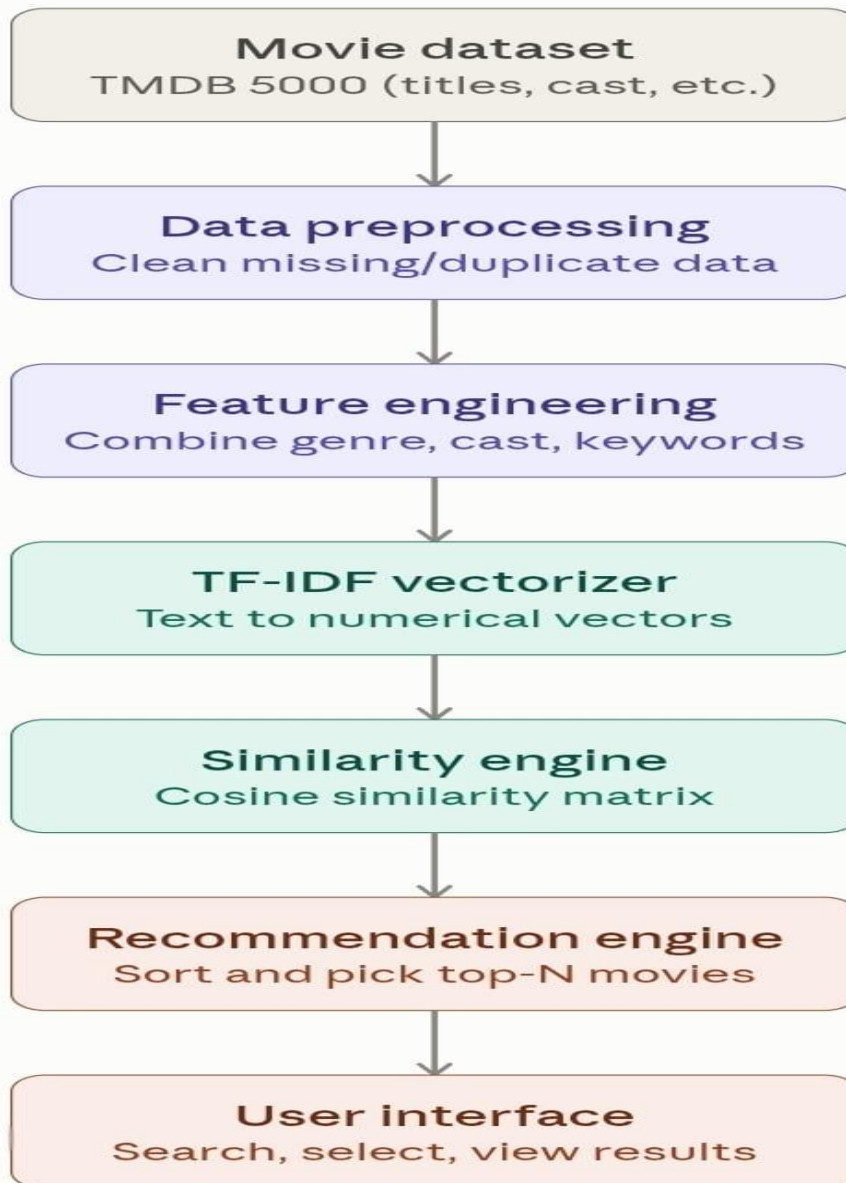
Project Goals and Milestones

- Movie-to-movie similarity recommendation.
- Search/select a movie and display similar movies.
- Display additional information such as genre, rating, release year, and poster when available.
- Support for larger movie metadata datasets.
- Future integration with user profiles and ratings.

Project Approach

1. Collect the movie dataset.
2. Clean missing, duplicate, and inconsistent data.
3. Select useful features such as genre, keywords, overview, cast, and director.
4. Combine selected textual features into one feature field for every movie.
5. Convert the combined text into numerical vectors using TF-IDF.
6. Calculate pairwise movie similarity using Cosine Similarity.
7. Accept a movie selection from the user.
8. Find the selected movie's similarity scores with other movies.
9. Sort the scores in descending order.
10. Display the top recommended movies.

System Architecture (High Level Diagram)



Project Outcome / Deliverables

This project will deliver a functional movie recommendation system, including: a web/mobile interface for browsing, rating, and receiving recommendations; a backend API for authentication and data handling; a recommendation engine using collaborative and/or content-based filtering; a movie database sourced from a public API (e.g., TMDB); a user data store tracking ratings and watch history; and supporting documentation. The final system will take a user's preferences and return personalized, ranked movie recommendations, evaluated using basic metrics such as precision or recall.

Assumptions

It is assumed that a reliable public movie dataset (e.g., TMDb, MovieLens) is available and sufficient for training and testing the recommendation model. It is assumed users will provide at least minimal interaction data (ratings or viewing history) for personalized recommendations to be effective; new users without history will receive generic/popularity-based recommendations (cold start). The system assumes stable internet connectivity for API-based data retrieval and does not account for real-time streaming licensing or regional content availability restrictions.

References

- MovieLens Dataset, GroupLens Research, University of Minnesota.
<https://grouplens.org/datasets/movielens/>
- Ricci, F., Rokach, L., & Shapira, B. (2015). *Recommender Systems Handbook*. Springer.